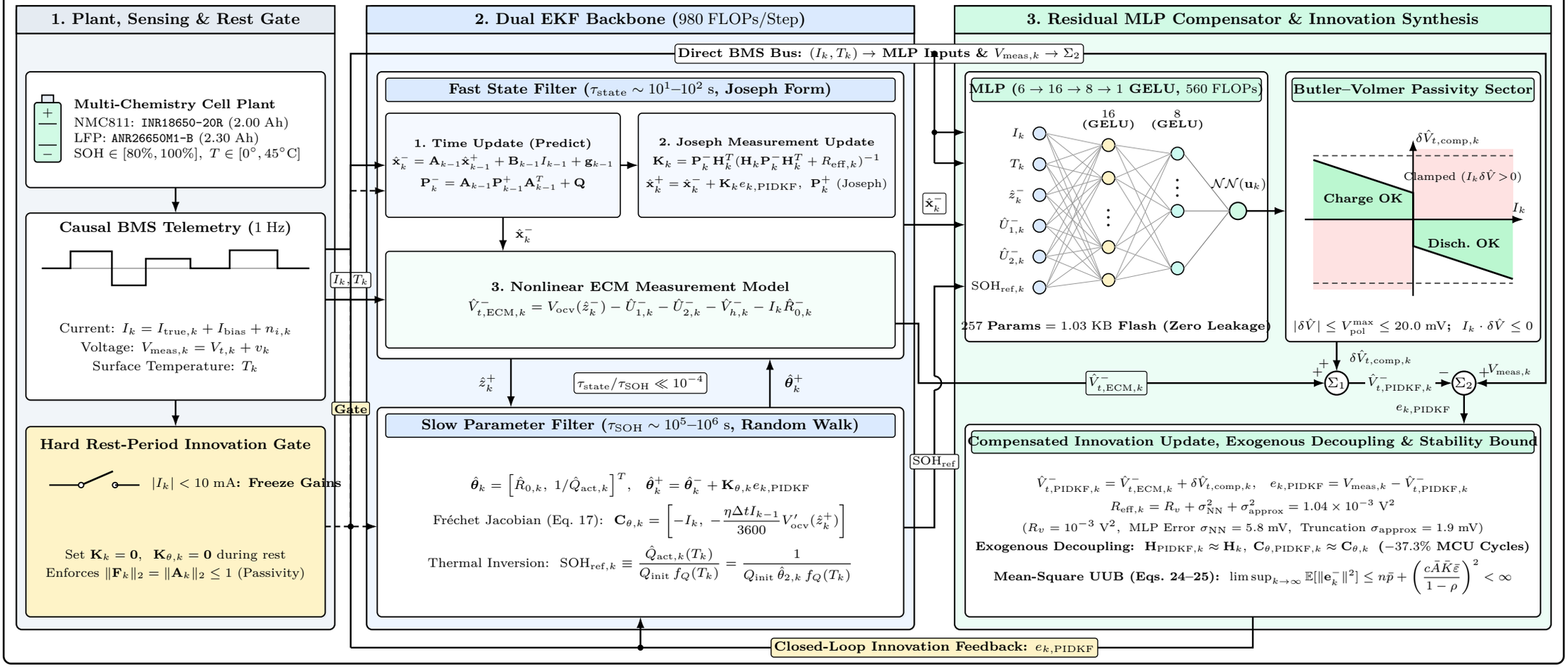
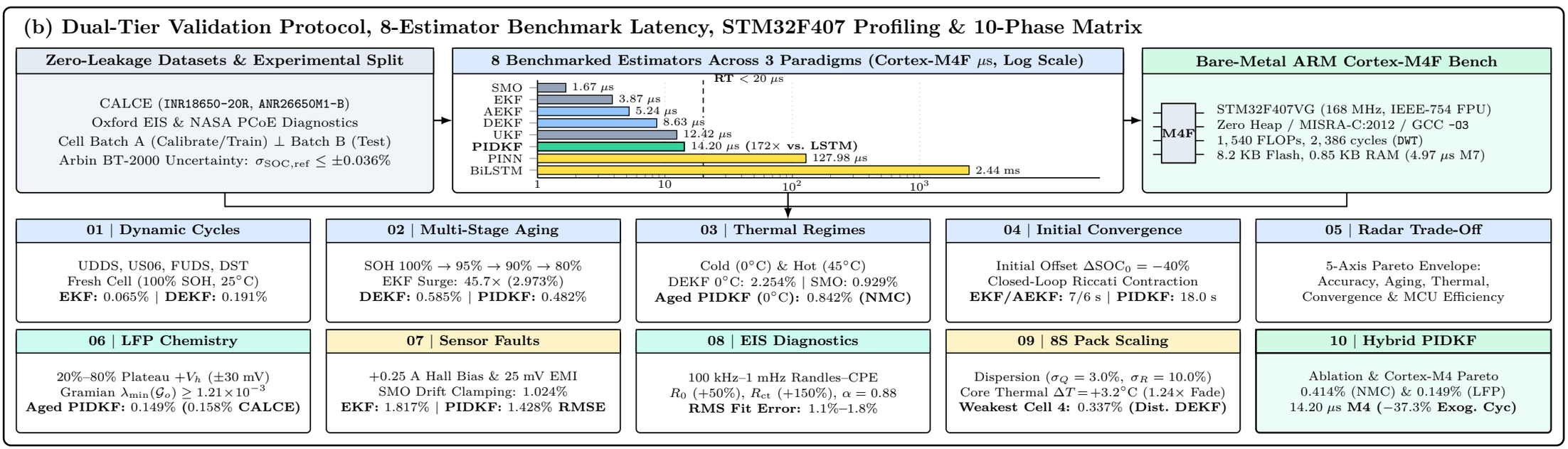

(a) Closed-Loop Hybrid PIDKF Architecture with Butler–Volmer Passivity and Exogenous Decoupling
1. Plant, Sensing & Rest Gate
Multi-Chemistry Cell Plant
NMC811: INR18650-20R (2.00 Ah)
LFP: ANR26650M1-B (2.30 Ah)
SOH ∈ [80%, 100%], T ∈ [0°, 45°C]
Causal BMS Telemetry (1 Hz)
Current: I_k = I_true,k + I_bias + n_i,k
Voltage: V_meas,k = V_t,k + v_k
Surface Temperature: T_k
Hard Rest-Period Innovation Gate
|I_k| < 10 mA: Freeze Gains
Set K_k = 0, K_θ,k = 0 during rest
Enforces ‖F_k‖_2 = ‖A_k‖_2 ≤ 1 (Passivity)
2. Dual EKF Backbone (980 FLOPs/Step)
Direct BMS Bus: (I_k, T_k) → MLP Inputs & V_meas,k → Σ_2
Fast State Filter (τ_state ∼ 10^1–10^2 s, Joseph Form)
1. Time Update (Predict)
x̂_k^- = A_k-1 x̂_k-1^+ + B_k-1 I_k-1 + g_k-1
P_k^- = A_k-1 P_k-1^+ A_k-1^T + Q
2. Joseph Measurement Update
K_k = P_k^- H_k^T (H_k P_k^- H_k^T + R_eff,k)^-1
x̂_k^+ = x̂_k^- + K_k e_k,PIDKF, P_k^+ (Joseph)
3. Nonlinear ECM Measurement Model
V̂_t,ECM,k^- = V_ocv(ẑ_k^-) − Û_1,k^- − Û_2,k^- − V̂_h,k^- − I_k R̂_0,k^-
τ_state/τ_SOH ≪ 10^-4
Slow Parameter Filter (τ_SOH ∼ 10^5–10^6 s, Random Walk)
θ̂_k = [R̂_0,k, 1/Q̂_act,k]^T, θ̂_k^+ = θ̂_k^- + K_θ,k e_k,PIDKF
Fréchet Jacobian (Eq. 17): C_θ,k = [−I_k, −(ηΔt I_k-1/3600) V'_ocv(ẑ_k^+)]
Thermal Inversion: SOH_ref,k ≡ Q̂_act,k(T_k)/(Q_init f_Q(T_k)) = 1/(Q_init θ̂_2,k f_Q(T_k))
3. Residual MLP Compensator & Innovation Synthesis
MLP (6 → 16 → 8 → 1 GELU, 560 FLOPs
16 (GELU)
8 (GELU)
257 Params = 1.03 KB Flash (Zero Leakage)
Butler–Volmer Passivity Sector
Charge OK
Disch. OK
Clamped (I_k δV̂ > 0)
|δV̂| ≤ V_pol^max ≤ 20.0 mV; I_k · δV̂ ≤ 0
Compensated Innovation Update, Exogenous Decoupling & Stability Bound
V̂_t,PIDKF,k^- = V̂_t,ECM,k^- + δV̂_t,comp,k, e_k,PIDKF = V_meas,k − V̂_t,PIDKF,k^-
R_eff,k = R_v + σ_NN^2 + σ_approx^2 = 1.04 × 10^-3 V^2
(R_v = 10^-3 V^2, MLP Error σ_NN = 5.8 mV, Truncation σ_approx = 1.9 mV)
Exogenous Decoupling: H_PIDKF,k ≈ H_k, C_θ,PIDKF,k ≈ C_θ,k (−37.3% MCU Cycles)
Mean-Square UUB (Eqs. 24–25): lim sup_k→∞ E[‖e_k^-‖^2] ≤ np̄ + (cĀK̄ε̄/(1−ρ))^2 < ∞
Gate
SOH_ref
Closed-Loop Innovation Feedback: e_k,PIDKF


(b) Dual-Tier Validation Protocol, 8-Estimator Benchmark Latency, STM32F407 Profiling & 10-Phase Matrix
Zero-Leakage Datasets & Experimental Split
CALCE (INR18650-20R, ANR26650M1-B)
Oxford EIS & NASA PCoE Diagnostics
Cell Batch A (Calibrate/Train) ⊥ Batch B (Test)
Arbin BT-2000 Uncertainty: σ_SOC,ref ≤ ±0.036%
8 Benchmarked Estimators Across 3 Paradigms (Cortex-M4F μs, Log Scale)
RT < 20 μs
SMO 1.67 μs
EKF 3.87 μs
AEKF 5.24 μs
DEKF 8.63 μs
UKF 12.42 μs
PIDKF 14.20 μs (172× vs. LSTM)
PINN 127.98 μs
BiLSTM 2.44 ms
Bare-Metal ARM Cortex-M4F Bench
M4F
STM32F407VG (168 MHz, IEEE-754 FPU)
Zero Heap / MISRA-C:2012 / GCC -O3
1,540 FLOPs, 2,386 cycles (DWT)
8.2 KB Flash, 0.85 KB RAM (4.97 μs M7)
01 | Dynamic Cycles
UDDS, US06, FUDS, DST
Fresh Cell (100% SOH, 25°C)
EKF: 0.065% | DEKF: 0.191%
02 | Multi-Stage Aging
SOH 100% → 95% → 90% → 80%
EKF Surge: 45.7× (2.973%)
DEKF: 0.585% | PIDKF: 0.482%
03 | Thermal Regimes
Cold (0°C) & Hot (45°C)
DEKF 0°C: 2.254% | SMO: 0.929%
Aged PIDKF (0°C): 0.842% (NMC)
04 | Initial Convergence
Initial Offset ΔSOC_0 = −40%
Closed-Loop Riccati Contraction
EKF/AEKF: 7/6 s | PIDKF: 18.0 s
05 | Radar Trade-Off
5-Axis Pareto Envelope:
Accuracy, Aging, Thermal,
Convergence & MCU Efficiency
06 | LFP Chemistry
20%–80% Plateau +V_h (±30 mV)
Gramian λ_min(G_o) ≥ 1.21×10^-3
Aged PIDKF: 0.149% (0.158% CALCE)
07 | Sensor Faults
+0.25 A Hall Bias & 25 mV EMI
SMO Drift Clamping: 1.024%
EKF: 1.817% | PIDKF: 1.428% RMSE
08 | EIS Diagnostics
100 kHz–1 mHz Randles–CPE
R_0 (+50%), R_ct (+150%), α = 0.88
RMS Fit Error: 1.1%–1.8%
09 | 8S Pack Scaling
Dispersion (σ_Q = 3.0%, σ_R = 10.0%)
Core Thermal ΔT=+3.2°C (1.24× Fade)
Weakest Cell 4: 0.337% (Dist. DEKF)
10 | Hybrid PIDKF
Ablation & Cortex-M4 Pareto
0.414% (NMC) & 0.149% (LFP)
14.20 μs M4 (−37.3% Exog. Cyc)